

The Alpha Factory

A strategy factory and a four-layer validation gate

Clarion Capital take-home · quantitative trader

Pre-registered criteria SHA-256 8e35cb...d5d8eab; full report in `report/report.pdf`

Verdict: 48 candidates in, 0 survivors out. A factory builds 48 strategies from two opposed market hypotheses; a four-layer gate, run independently on all of them, admits none through all four. The strongest, a vol-targeted trend on ETH with an annualised Sharpe of 1.17, still fails: it breaches the drawdown cap in the 2020 crash, and its Deflated Sharpe is 0.35, far below the 0.95 bar. Nothing here is alpha, and that honest result is what the project defends. The brief rewards the machine and the reasoning, not survivors.

The factory

The factory builds 48 instances from two deliberately *opposed* families. `voltarget_mom` is vol-targeted momentum, a trend-following bet: it takes the sign of the return over a lookback and scales the position so it targets constant volatility, capped at $2\times$ leverage. `bollinger_revert` is filtered mean-reversion, a reversal bet: it fades a Bollinger z -score beyond a threshold, but only in calm regimes (a longer-horizon trend gate), and exits flat when price crosses back through its moving average. Because the two families are opposed by construction, their signal correlation is about -0.30 (versus about $+0.60$ between siblings of the same family): genuine diversification, not one bet dressed up as two.

Each family exposes at most two free parameters, the real hypothesis knobs (horizon and sensitivity): for the trend family the lookback and the volatility window, for the reversion family the lookback and the z -threshold. Everything else, the volatility baseline (about three months), the leverage cap, the trend gate and the moving-average exit, is fixed by domain reasoning, because optimising a risk limit or a reference scale is textbook overfitting. With six parameter settings per family across four markets (S&P 500, USD/JPY, gold, ETH), the population is $2 \times 6 \times 4 = 48$. A vectorised backtester evaluates each instance with a one-bar execution lag, verified by a truncation test (hiding the future leaves every past signal bit-identical), and nets realistic costs from each market's own measured half-spread.

The four-layer gate

Four independent tests judge the 48 instances. They run *in parallel* on the full population, and the survival funnel is their *intersection*, not a sequential filter that would hide what later layers reject; the intersection is what exposes that the Deflated Sharpe alone rejects all 48.

- **L1, out-of-sample and robustness.** Requires a positive out-of-sample edge and stability under $\pm 20\%$ parameter perturbations. With parameters fixed by design the in-sample split trains nothing, so the neighbour-robustness checks are the real anti-overfit teeth.
- **L2, walk-forward.** Re-tests each instance across consecutive out-of-window blocks and passes only those whose aggregated out-of-window Sharpe clears 0.50 and stays positive in at least 60% of windows.
- **L3, stress replay.** Requires surviving the 2020 COVID crash *and* each instance's own worst 21-day windows within a drawdown cap, plus a volatility-only $2\times$ synthetic shock.
- **L4, significance.** Demands a block-permutation p -value below 0.05 (which preserves serial structure rather than assuming i.i.d. returns) *and* a Deflated Sharpe of at least 0.95.

Layer	What it tests	Pass	Dominant failure
L1	OoS edge + $\pm 20\%$ robustness	4 / 48	OoS Sharpe below bar
L2	Walk-forward stability	6 / 48	out-of-window Sharpe/return low
L3	Stress replay (COVID, own worst, $2\times$ vol)	8 / 48	blows up in the COVID window
L4	Block-permutation + Deflated Sharpe	0 / 48	fails the Deflated Sharpe
Intersection (all four layers)		0 / 48	

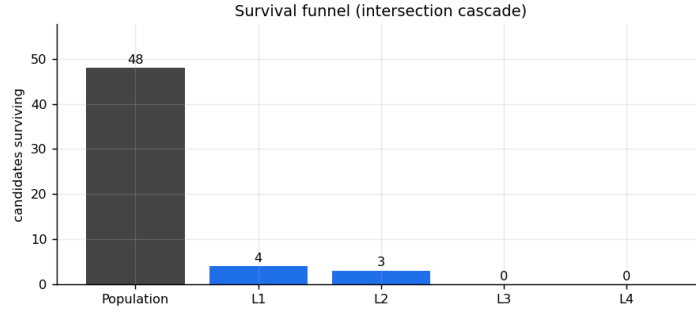


Figure 1: The intersection cascade: 48 candidates in, 0 through all four independent layers.

Why nothing survives

The binding constraint is L4's **Deflated Sharpe Ratio**, which corrects a Sharpe for the number of strategies tried. Searching 48 configurations inflates the best one by selection: the luckiest of 48 zero-skill trials would be expected to score about 1.32 annualised (the dispersion of the 48 Sharpes times an extreme-value multiplier that grows with the trial count). The strongest candidate, the ETH trend, earns a gross Sharpe of 1.38 and 1.17 after costs, with a block-permutation p -value of 0.001, so its edge is *not* a serial-structure artifact; yet its 1.17 sits *below* that luck hurdle, giving a Deflated Sharpe of only 0.35 against the 0.95 bar. Even crediting the redundancy among correlated trials, an effective count of about 12 rather than 48, lifts it only to 0.69. The same candidate also breaches the drawdown cap in 2020 (about 26% in the COVID window, 47% in its own worst window), so L3 rejects it independently. Generating *more* candidates cannot manufacture a survivor: each one raises the bar for all of them.

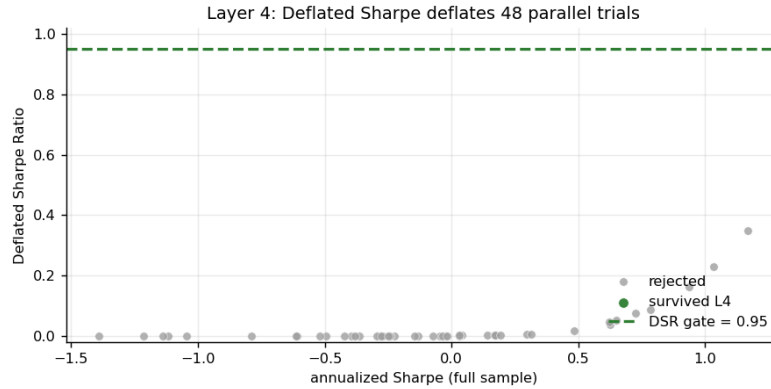


Figure 2: Deflated Sharpe versus net Sharpe across the 48 instances. None clears the 0.95 bar; the best reaches 0.35.

Where the costs bite

The two families fail for different reasons. The reversion grid is *low* turnover (about 151 trades per year, since it sits flat between trades) but its signal is weak: it is negative even gross, so costs are not what kills it. The trend grid is *high* turnover (about 713 trades per year, because it re-scales the position every bar to hold volatility constant) and is exactly where costs bite, the median trend instance falling from a gross Sharpe of 0.59 to 0.11 net. A naive read would blame costs for both; only the trend family is genuinely cost-sensitive.

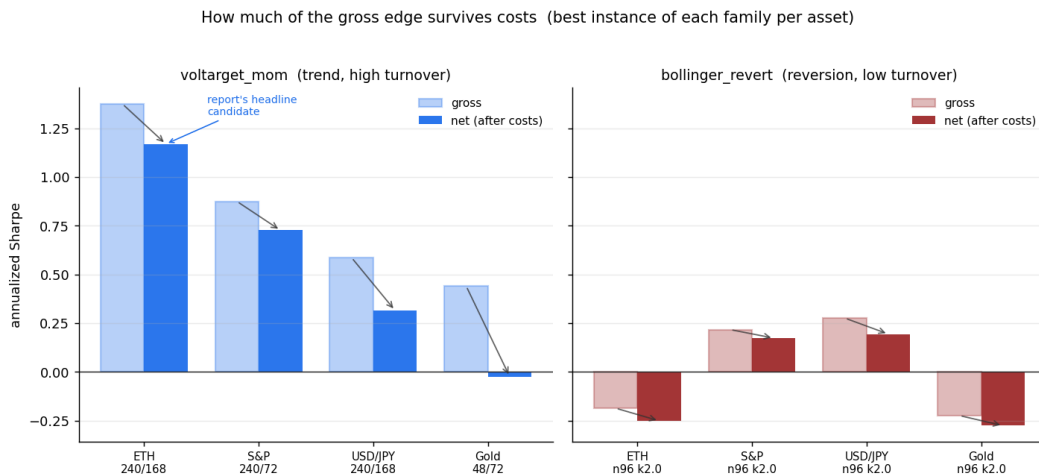


Figure 3: Gross versus net Sharpe for the best instance per market, by family. Costs barely move the low-turnover reversion grid but materially erode the high-turnover trend grid.

Honesty and reproducibility

Every threshold was fixed in `config/gate_criteria.yaml` before the gate ran, and `run.py` prints its SHA-256. The hash was identical across all four strategy iterations; it changed exactly once, for a documented L2 redesign (from a *relative* selection-frequency verdict, where instances competed, to an *independent* per-instance test). Three further refinements are catalogued in the repository (L1’s soft-guard explanation, L3’s move to each instance’s own worst windows, L4’s upgrade from i.i.d. to block permutation). None moved the verdict, which stayed at 0 survivors throughout, the strongest evidence that the criteria were never loosened to admit a winner. The pipeline is deterministic (fixed seed) and reproduces $48 \rightarrow 0$ from the raw data, byte-identically, with `python run.py`.

Limitations

The cuts are deliberate simplicity, not laxity: two families and coarse parameter grids (a richer factory is the obvious next step), a single H1 timeframe, a cost model based on the mean measured half-spread without slippage or market impact, and a conservative raw trial count of 48 in the Deflated Sharpe (the effective count of about 12 is the more principled figure, reported alongside). Each is a candidate for the next iteration, a richer factory, never a more permissive gate.